GROW computing eight week plan

Scratch maze project | Eight lessons of 40 minutes

Teach one shared computing goal through a predictable GROW chassis, then adjust prompts, task size and challenge to the pupil's current access and understanding. Weeks 1 and 2 include three usable task routes in the accompanying packs.

AQA 71638 Programming with Scratch unit 6

The current unit is Level One. All six outcomes remain the completion target. Planned teaching time is 320 minutes; these are local estimates. Agree additional sessions at the Week 2 and Week 6 reviews if necessary.

Week	Shared focus	Assessment destination
1	Get a result	Foundation practice
2	Control a character	Foundation practice
3	Build the world	Outcome 1
4	Make controls reliable	Outcome 2
5	Make a rule work	Outcome 3
6	Finish and restart	Outcome 4
7	Add one challenge	Outcome 5
8	Test and improve	Outcome 6 and complete review

Choose the route from observed need

At We Do 2 · Task, ask the pupil to show the first action and explain or point to its likely effect. Offer Step by step if locating, remembering or reading the step prevents participation. Use Main task when the pupil can begin with a reference. Offer Challenge when the pupil can complete and explain the shared task. Recheck after a success and withdraw one prompt.

Evidence stays individual

Record what the pupil created, changed, tested and explained, plus the support used. Supplied practice, teacher models and recovery code do not prove pupil authorship. Directing an adult to operate a device can support access; record the method and agree acceptable evidence with the UAS lead. Do not assume an adult's action proves the pupil created code.

Full teaching packs are supplied for Weeks 1 and 2. Weeks 3-8 below specify the teaching, access resources to prepare and assessment opportunities. The separate scheme of work retains all continuation and alternative routes.

The GROW lesson chassis

The same eight stages in every 40-minute lesson

Stage and time	What pupils and adults do
Arrival 0-3 minutes	Settle, open the right file and see one first action. Offer a quiet help signal and a clear return point after a break.
Starter 3-6 minutes	Retrieve one idea. Everyone predicts by speaking, pointing, showing a card or writing. Do not reveal the answer first.
I Do 6-11 minutes	Model one small action. Say what you notice and why you choose the block. Keep the relevant code visible.
We Do 11-18 minutes	Pupils predict, direct or perform the next step. Run the test, compare the result and explain one change together.
I Do 2 18-21 minutes	Model the next small skill or a repair. Connect it explicitly to the shared test.
We Do 2 · Task 21-25 minutes	Try the first task step together. Check understanding and choose a route before pupils work more independently.
Independent Task 25-35 minutes	Use a flexible route. Fade a prompt after success, change route when needed and observe each pupil’s own contribution.
Exit 35-40 minutes	Save, reopen and show one result. Record pupil voice and give an adult response or a specific follow-up time.

A shared goal with flexible routes

Step by step provides more modelling, visual prompts and smaller actions. Main task gives the shared task with a reference available. Challenge adds a constraint, explanation or harder test after the shared goal. Routes describe today’s support, not a fixed pupil group. A pupil can change route within or between lessons.

Use short sentences pitched around reading ages 7-11, one action at a time and visible code blocks. Offer spoken, pointed or exact-word scribed responses, quiet help and a visible return step after regulation time. If sharing a device, swap keyboard roles and observe each pupil.

Week 1 Get a result

GROW chassis | 40 minutes | Select and review the route during the lesson

Shared goal: I can run a program, change one value and save my work.

Words: sprite, event, script, value.

Stage and time	Teaching and pupil action
Arrival 0-3	Settle. Open W01_Start. Point to Courier and the green flag.
Starter 3-6	Predict which travels further from x0: move 40 or move 100.
I Do 6-11	Model event, start, direction, move and message. Run twice.
We Do 11-18	Predict/run 40 then 100. Keep the start and direction fixed. Explain why two reset runs both end at 100.
I Do 2 18-21	Model Save to your computer and loading the saved .sb3.
We Do 2 · Task 21-25	From x0, choose a move value for the marker at x120. Predict, run and check together.
Independent Task 25-35	Use the chosen route. Change the value and message, test twice and show the individual change.
Exit 35-40	Save W01_MyCode, reopen and run. Share one result and a voice response.

Step by step: Use the Step by Step booklet. Point on the number track, choose 40 or 100, then change only the move value to reach 120. Read each step aloud if needed; rehearse saving with a file picture.

Main task: Use the main booklet to predict 40 and 100, reach 120, choose a short message, test twice and save/reopen. Explain one changed value.

Challenge: After Main task, reach 160, test twice and explain the reset. If time permits, use the earlier HTML model to predict and compare two 100-step runs with and without reset.

Voice and response: Choose a message or a way of getting help. Show the changed program or access step. Use “What I said, and what it changed”.

Evidence and review: Foundation practice for later movement and testing. No complete 71638 outcome is claimed from this lesson alone.

Resources: Week 1 HTML or slides, W01_Start.sb3, chosen pupil booklet, teacher guide and the approved .sb3 save folder.

Week 2 Control a character

GROW chassis | 40 minutes | Select and review the route during the lesson

Shared goal: I can use key events to move a sprite and fix a direction mistake.

Words: input, x, y, debug.

Stage and time	Teaching and pupil action
Arrival 0-3	Retrieve the green-flag event. Open W02_Start with reset only.
Starter 3-6	Watch a left key move right. Predict the cause before seeing a fix.
I Do 6-11	Build right/change x by 10 and left/change x by -10 in a demo.
We Do 11-18	Repair the wrong sign together. Predict/test left then right; reset and trace right/right/left/left.
I Do 2 18-21	Model up/down using y. Positive y goes up.
We Do 2 · Task 21-25	Plan and test the beacon route to (20,20). Use (20,0) while consolidating the horizontal axis.
Independent Task 25-35	Build the pupil controls using the chosen route. Test, explain one script and record a mission idea.
Exit 35-40	Save W02_MyControls, reopen and test. Record a response to the mission idea.

Step by step: Use the Step by Step booklet and one axis. Build right, test, then build left using the block reference. Point to the minus sign. Return to any unfinished second script.

Main task: Build both horizontal scripts from W02_Start, test right/right/left/left from reset, explain one input/action link and save/reopen.

Challenge: After Main task, add up/down. Reach (20,20) in two orders and explain why both work here. A wrong-axis or wrong-sign diagnosis is an optional extra if time permits.

Voice and response: Choose a mission and suggest one feature. Teacher adopts, adapts or defers it for Week 3 or Week 7, with a reason. Use “What I said, and what it changed”.

Evidence and review: Practice towards movement and testing. Confirm outcome 2 in the main maze at Week 4. Do not sign off the whole unit or infer pupil authorship from recovery code.

Readiness: one constructed and explained key script can show readiness for the next build, but both horizontal scripts remain the Week 2 task. Return to any gap; agree extra foundation time before Week 3 if needed.

Resources: Week 2 HTML or slides, W02_Start.sb3, chosen pupil booklet, teacher guide and the approved .sb3 save folder.

Week 3 Build the world

GROW chassis | 40 minutes | Select and review the route during the lesson

Shared goal: Create a playable layout with separate maze, player and finish sprites.

Words: sprite, backdrop, route, finish.

Stage and time	Teaching and pupil action
Arrival 0-3	Open a new project; retrieve the meaning of sprite.
Starter 3-6	Inspect a route that is too narrow.
I Do 6-11	Draw a maze as a sprite; leave a clear start and route.
We Do 11-18	Trace the route with a token; adjust one passage together.
I Do 2 18-21	Create and name separate Player and Finish sprites.
We Do 2 · Task 21-25	Check scale and place the player and finish together.
Independent Task 25-35	Pupils create their three sprites and a short playable layout.
Exit 35-40	Save W03_Maze.sb3 and identify all three sprites.

Step by step: Rehearse a short wide route with a token. Give three role cards: Maze, Player, Finish. Reveal one creation step at a time; pupils create all three sprites.

Main task: Create and name the three sprites. Draw a short maze, size Player to fit and place Finish at a reachable end.

Challenge: Keep the three sprites. Compare two player sizes or a narrow passage; justify and test a layout change that keeps the game playable.

Voice and response: Revisit the Week 2 decision. Ask which route or object should change; show the accepted layout change. Use “What I said, and what it changed”.

Evidence and review: Outcome 1: observe pupil creation and identification of all three sprites. A backdrop alone does not meet the maze-sprite requirement.

Prepare: the previous saved project, a separate teacher demo and the prompts described in the chosen route. Keep the next code step visible.

Week 4 Make controls reliable

GROW chassis | 40 minutes | Select and review the route during the lesson

Shared goal: Create and test all four arrow controls in the main maze.

Words: event, x, y, reset.

Stage and time	Teaching and pupil action
Arrival 0-3	Reopen the maze and select Player.
Starter 3-6	Predict the result of one wrong-sign control.
I Do 6-11	Model constructing horizontal key scripts in a separate example.
We Do 11-18	Build and test right/left in the main maze.
I Do 2 18-21	Model up/down and a green-flag start position.
We Do 2 · Task 21-25	Agree a four-direction test from the same start.
Independent Task 25-35	Pupils create all four controls and test them in their maze.
Exit 35-40	Save W04_Controls.sb3 and demonstrate all directions.

Step by step: Use four key/block cards. Build and test x controls first, then y. Cover completed reference rows. Give extra construction time if all four are not ready.

Main task: Create four arrow scripts in the maze, set a start position and test each direction from reset. Rebuild any previously inherited scripts.

Challenge: Predict how controls behave after changing sprite direction. Compare change x/y with move steps in a copy, then retain reliable controls.

Voice and response: Ask which control feels difficult. Agree and test a useful change in step size or practice support. Use “What I said, and what it changed”.

Evidence and review: Outcome 2: observe the pupil create and demonstrate arrow-key scripts. Revisit any inherited Week 2 code.

Prepare: the previous saved project, a separate teacher demo and the prompts described in the chosen route. Keep the next code step visible.

Week 5 Make a rule work

GROW chassis | 40 minutes | Select and review the route during the lesson

Shared goal: Use a condition to trigger a consequence on contact with the maze.

Words: condition, touching, consequence, loop.

Stage and time	Teaching and pupil action
Arrival 0-3	Open the working controls version.
Starter 3-6	Predict a wall contact with no consequence.
I Do 6-11	Model forever + if touching Maze then go to the start.
We Do 11-18	Trace the condition and consequence together.
I Do 2 18-21	Model a check attached to the wrong sprite and fix it.
We Do 2 · Task 21-25	Agree tests at two different walls.
Independent Task 25-35	Pupils construct their contact script and test it.
Exit 35-40	Save W05_Collision.sb3 and explain the consequence.

Step by step: Use contact/no-contact picture cards and a three-part prompt: keep checking, touching Maze?, return to start. Build the script in the pupil project and test two walls.

Main task: Create a repeated collision check with a clear consequence. Predict and test contact and no contact at two positions.

Challenge: Test a corner or narrow gap. Explain a stuck-player fault, change one cause and retest both that case and an ordinary wall.

Voice and response: Ask whether the contact response is clear and fair. Agree a suitable response while retaining a consequence. Use “What I said, and what it changed”.

Evidence and review: Outcome 3: observe construction and demonstrated contact response. Save before and after a repair when useful.

Prepare: the previous saved project, a separate teacher demo and the prompts described in the chosen route. Keep the next code step visible.

Week 6 Finish and restart

GROW chassis | 40 minutes | Select and review the route during the lesson

Shared goal: Create a visible winning response and restore a playable new run.

Words: win condition, state, message, restart.

Stage and time	Teaching and pupil action
Arrival 0-3	Open the collision version and reset.
Starter 3-6	Predict what happens at a finish with no win code.
I Do 6-11	Model a finish condition and visible win response.
We Do 11-18	Predict and run a complete route in the example.
I Do 2 18-21	Model a new green-flag run after winning.
We Do 2 · Task 21-25	Agree win and restart tests together.
Independent Task 25-35	Pupils construct a win response and test a fresh run.
Exit 35-40	Save W06_Win.sb3; review any missing core outcomes.

Step by step: Use a two-picture test: at Finish / away from Finish. Choose a short visible message. Build the win script, then use a save and restart strip.

Main task: Create a finish condition and visible win response. Test a win, then a fresh green-flag run with the wall rule still working.

Challenge: Investigate a repeated win message or a changed state that survives restart. Explain and test a repair without breaking collision checks.

Voice and response: Ask which win signal is clear and comfortable. Test the chosen message or visible change. Use “What I said, and what it changed”.

Evidence and review: Outcome 4: pupil creates and demonstrates the response. Review unfinished outcomes now.

Time decision: check the first complete run and outcomes 1-4. Agree extra construction or testing sessions now if the game is not ready for the required complexity feature.

Prepare: the previous saved project, a separate teacher demo and the prompts described in the chosen route. Keep the next code step visible.

Week 7 Add one challenge

GROW chassis | 40 minutes | Select and review the route during the lesson

Shared goal: Add one new mechanic that changes what the player must do.

Words: mechanic, obstacle, collect, test.

Stage and time	Teaching and pupil action
Arrival 0-3	Open the working game and save a new copy.
Starter 3-6	Compare a new background with a moving obstacle.
I Do 6-11	Model an obstacle moving across a route in a separate project.
We Do 11-18	Choose one feature and state its effect on play.
I Do 2 18-21	Model a contact consequence for the extra obstacle.
We Do 2 · Task 21-25	Predict an ordinary test and a difficult case.
Independent Task 25-35	Pupils build the feature and test the changed game.
Exit 35-40	Save W07_Challenge.sb3; explain the extra challenge.

Step by step: Choose one slow moving obstacle with a contact consequence. Use a movement route sketch and one-block-at-a-time prompts; pupil builds the added behaviour.

Main task: Choose an achievable moving obstacle or meaningful collectible. Build the mechanic and show how it changes a player action or decision.

Challenge: Tune the mechanic for a fair challenge. Predict an awkward case, compare two settings and justify the retained version with a test.

Voice and response: Revisit the Week 2 feature request. Agree an achievable mechanic and explain any adaptation. Use “What I said, and what it changed”.

Evidence and review: Outcome 5: demonstrate increased gameplay complexity. Decoration alone is insufficient.

Prepare: the previous saved project, a separate teacher demo and the prompts described in the chosen route. Keep the next code step visible.

Week 8 Test and improve

GROW chassis | 40 minutes | Select and review the route during the lesson

Shared goal: Test the game against clear expectations and explain an improvement.

Words: expected, actual, fault, retest.

Stage and time	Teaching and pupil action
Arrival 0-3	Load the latest game and prepare the test record.
Starter 3-6	Compare “it is broken” with a specific fault report.
I Do 6-11	Model expected versus actual results from a known start.
We Do 11-18	Agree tests for controls, walls, finish, extra feature and reset.
I Do 2 18-21	Model one small repair and retest the same sequence.
We Do 2 · Task 21-25	Choose a peer or teacher tester and rehearse feedback.
Independent Task 25-35	Pupils test, improve and retest their games.
Exit 35-40	Save W08_Final.sb3 and review the six-outcome record.

Step by step: Use five pictured test prompts: arrows, walls, finish, extra feature, restart. Say or point to expected/actual results while an adult records exact words.

Main task: Choose a peer or teacher tester. Run the five checks, record a precise fault, make one useful improvement and retest the same steps.

Challenge: Add an edge-case test from a known start. Explain what an ordinary run missed and check the repair against earlier successful tests.

Voice and response: Ask what a tester’s feedback changed. Identify the actual improvement or explain why a suggestion was deferred. Use “What I said, and what it changed”.

Evidence and review: Outcome 6: observe testing. Review every outcome individually before any award claim.

Prepare: the previous saved project, a separate teacher demo and the prompts described in the chosen route. Keep the next code step visible.